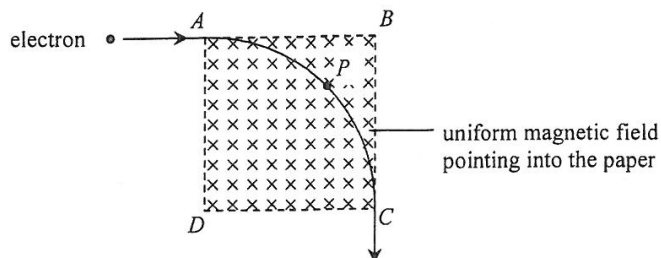


- *4. An electron moving with speed $1.2 \times 10^7 \text{ m s}^{-1}$ enters a square region $ABCD$ with a uniform magnetic field of 0.01 T pointing into the paper as shown in Figure 4.1. The electron describes a quarter circle from A to C and it emerges from C with the same speed. Neglect the effects of gravity.

Figure 4.1



- (a) (i) Find the magnitude of the magnetic force acting on the electron at point P on its path. (2 marks)

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- (ii) Indicate in Figure 4.1 the direction of the electron's acceleration at the point P . (1 mark)

- (b) Although the electron accelerates due to the magnetic force, explain why it emerges from the magnetic field with the same speed. (2 marks)

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- (c) Deduce the speed of the electron when entering the magnetic field such that it would describe a semi-circle from A to D instead. (2 marks)

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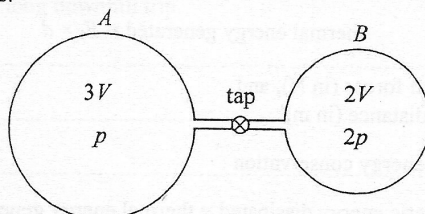
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- *2. Two vessels, A and B , of volumes $3V$ and $2V$ respectively are connected by a narrow tube with a tap as shown in Figure 2.1. Initially the tap is closed and both vessels are at the same temperature. Vessel A contains helium gas at a pressure p while vessel B contains 0.8 mol helium gas at a pressure $2p$. Assume that helium gas can be taken as an ideal gas.

Figure 2.1



- (a) Deduce the amount of helium gas (in mol) in vessel A . (2 marks)

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- (b) Now the tap is open and a steady state is reached. Assume that the temperature remains unchanged.

- (i) Find the gas pressure inside the vessels in terms of p . (2 marks)

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- (ii) Account for the pressure change of the gas in vessel A using kinetic theory. (2 marks)

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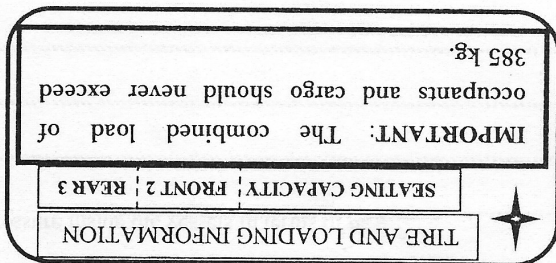
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- (a) Referring to the passage, explain why there should be a maximum load limit for a vehicle (see the vehicle label example below). (2 marks)

3. Read the following passage about braking in vehicles and answer the questions that follow.

Mechanical braking is the most common braking method in vehicles. In this method, wheels of vehicles are fitted with brake pads which apply frictional forces that inhibit the motion of the wheels. Frictional braking results in a conversion of the vehicle's kinetic energy to thermal energy which then dissipates to the atmosphere in the form of waste heat.

The thermal energy (in J) generated during braking is approximately given by :

$$\text{thermal energy generated} = F_f \times d$$

where F_f is the frictional forces (in N), and d is the braking distance (in m).

Applying the principle of energy conservation :

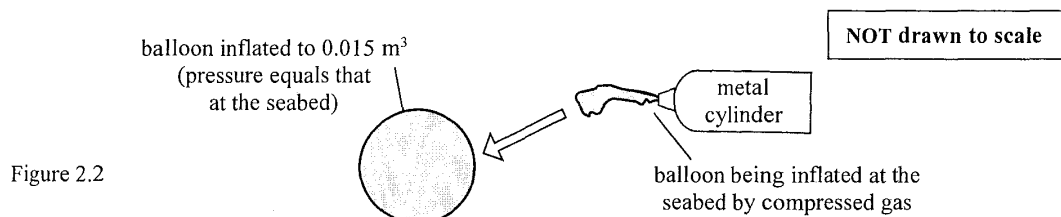
$$\text{kinetic energy dissipated} = \text{thermal energy generated}$$

$$\frac{1}{2}mv^2 = F_f \times d$$

From this equation it can be seen that increasing the velocity v or mass m of an object means the applied frictional forces must be increased in order to bring the object to a stop within the same distance.

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*(c) The diver then inflates identical balloons each to a volume of 0.015 m^3 by using the cylinder of compressed gas at the seabed. Assume that the balloons are inflated slowly so that the temperature of the gas remains unchanged and the final pressure in the balloon equals that at the seabed.



- (i) Show that the gas pressure in the cylinder decreases by 5.0 atm after inflating one balloon. (2 marks)

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- (ii) Hence, find the total number of balloons that the diver can inflate completely. (2 marks)

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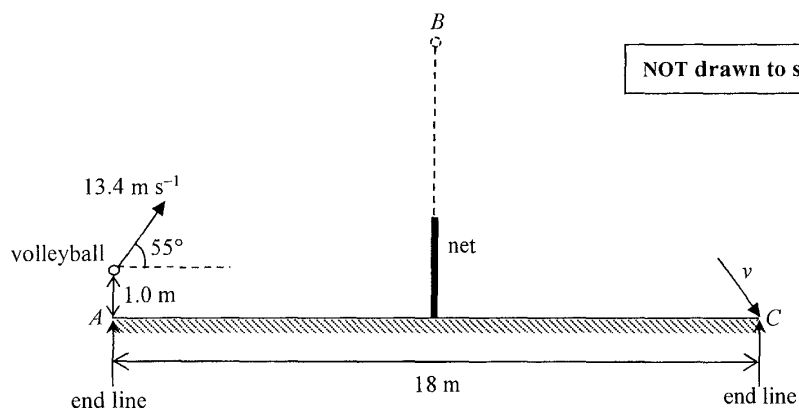
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3. A volleyball player serves by hitting the ball from rest at a height of 1.0 m above the end line of the court. The initial speed of the ball is 13.4 m s^{-1} making an angle of 55° with the horizontal. It moves in a vertical plane perpendicular to the end line and finally reaches point C on the opposite end line as shown in Figure 3.1. Neglect the size of the ball and air resistance. ($g = 9.81 \text{ m s}^{-2}$)

Figure 3.1



- (a) (i) The mass of the volleyball is 0.22 kg. Find the work done on the ball by the player. (2 marks)

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- (ii) Determine the speed v with which the ball hits point C on the ground. (2 marks)

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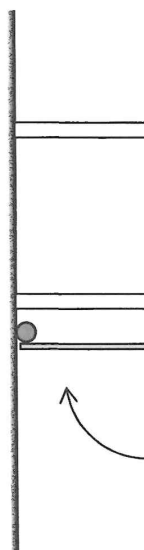
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Figure 3.2



- (b) (i) If the mass of the ball is 0.05 kg and the collision time is 0.008 s , find the magnitude of the average force acting on the ball by the rod. (2 marks)

- (ii) Explain why the momentum of the ball is not conserved before and after the collision. (1 mark)

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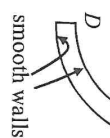
table surface



Figure 4.1(a)

A small sphere is projected along the track with speed u at A . It moves along the track and passes B at 4.0 m s^{-1} . Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

Figure 4.1(b)



(a) By using the law of conservation of energy, find u .

(2 marks)

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Answer **ALL** questions. Parts marked with “*” involve knowledge of the extension component. Write your answers in the spaces provided.

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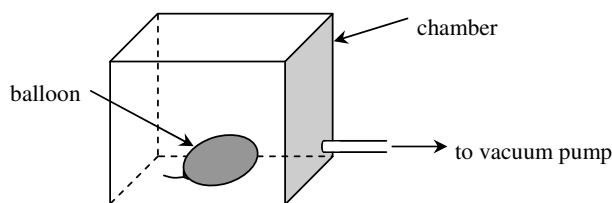


Figure 1.1

A balloon containing 0.01 m^3 of gas at a pressure of 100 kPa is placed inside a chamber. Air is slowly pumped out from the chamber while the temperature remains unchanged.

- *(a) Explain, in terms of molecular motion, how the gas inside the balloon exerts a pressure on its inner surface. (2 marks)

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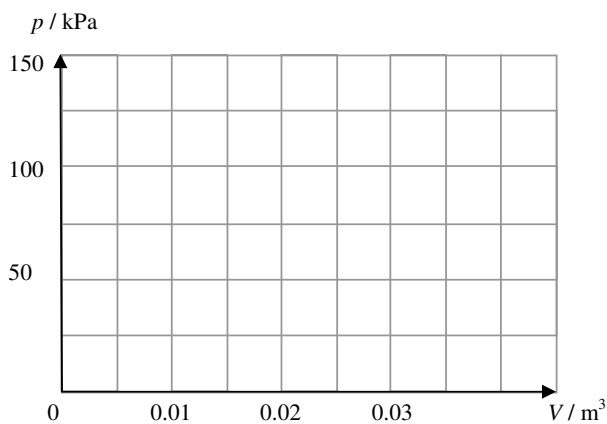
- *(b) Find the final pressure inside the balloon when its volume is doubled. (2 marks)

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- *(c) Sketch a graph to show the relationship between the pressure p inside the balloon and the volume V of the balloon. (2 marks)



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